SE(3)-WAM: What Should a Robot Remember and Predict for Dynamic Manipulation? Supplementary Material

Anonymous CVPR submission

Paper ID *****

This supplement provides the complete experimental protocol, implementation, condition definitions, and measured results. Planned comparisons are identified where their outcomes remain unavailable. The same appendix sources are included in the standalone supplement and the complete English reading edition.

Appendix	Contents	Main
A	Tasks, data, information, scoring	Sec. 3
B	Implementation and system budgets	Secs. 4–5
C	History conditions and results	Sec. 5.4
D	Future targets, statistics, combinations	Secs. 5.2–5.3
E	Backbone and information flow	Secs. 4.4, 5.6
F	Pretraining, deployment, external tests	Sec. 5.6
G	Failures and qualitative evidence	Sec. 5.5

A. Tasks, Data, and Evaluation

A.1. The Reported Fourteen-Task Suite

The completed studies use the migrated DB0 v0.6 demonstration pool and its fourteen-task evaluation suite. Table S1 lists the task identifiers used in the result arrays. These identifiers are retained so that results cannot be silently pooled into a changed task taxonomy. The task-balanced score includes every task, including zero-success conditions.

The common training pool has 11,200 episodes, 800 per task. The internal representation screen uses 350 validation episodes; adapted systems retain their own validation settings (including 1,400 for DreamVLA/OpenWAM). The future screen samples 160,416 chunks per epoch. Stored episode count and effective chunk exposure measure different aspects of training. Later fresh-data production retains its own data identity and does not retroactively change the source of these results. Motion-only P0 prediction has no corresponding action-learning score in the fourteen-task table.

Success is read from the native task evaluator after its ordered physical requirements. A final pose or incidental contact is not substituted for the evaluator outcome. Native failure categories are boundary, safety, task violation, and timeout. These categorical outcomes are useful for localization of failures; the word safety is an evaluator category rather than a certified real-robot safety guarantee. The exact numerical event thresholds and evaluator configuration for an independently distributable release remain **[release configuration to attach]**. Historical grasp thresholds from earlier manuscripts are not substituted for those actual settings.

A.2. Partitions and Statistical Interpretation

The 140-scene development set has ten scenes per task. Its disjoint complement has ninety per task, totaling 1,260. The future screen reports the two partitions separately. The system-reference study also keeps them separate. The combination screen explicitly selects on all 1,400 scenes while retaining both component counts. Thus its selection score includes development data, and the 1,260 complement is

Table S1. Task identifiers and the interaction they exercise. These are the reported evaluation tasks, not a count of new environments introduced by this manuscript.

Identifier	Interaction
grasp	Reference object grasping
t1_belt	Moving conveyor target
t1_occ	Moving target with physical occlusion
t1_so3	Orientation-sensitive target grasping
t1_xyz	Translating target grasping
t2_se3	Placement into a rigidly moving receptacle
t2_trans	Placement into a translating receptacle
t3_full	Full constrained-passage task
t3_phase	Phase-dependent passage
t4_can	Roll/slip can capture
t4_slider	Sliding-object capture
t4_sphere	Rolling-sphere capture
t4_sphere_tabletop	Tabletop sphere capture
t5_replan	Post-impact replanning

now part of the project's development evidence. Neither membership disjointness nor a different rollout seed makes it a final blind test.

The future study uses each condition's own fourth-epoch endpoint and one training seed. Across arms, the scene, source seed, and inference seed are paired. There are 18,900 accepted outcomes, 1,260 per arm; missing scenes and infrastructure-error exclusions are zero. Inference seeds span 20261058–20262317 for the designated selection indices. Matching an inference seed specifies rollout randomness but does not imply that unlike training graphs consumed identical random-number sequences.

For two policies on the same scenes, let b count candidate successes/reference failures and c candidate failures/reference successes. Then

$$\Delta\mathrm{SR} = 100(b-c)/1260 \tag{S1}$$

in percentage points. The report uses the exact two-sided McNemar test on discordant pairs. Fourteen comparisons to P00 and thirteen to P03 form the twenty-seven-comparison family; adjusted values use $\min(1, 27p)$. The 378 task-level comparisons remain exploratory. No confidence interval is invented for a source that reports none, and scenario-level tests do not estimate training-seed variation.

A.3. Inputs and Time

The completed future screen uses one current fixed-camera RGB view, exact depth/camera geometry, robot-state history, and privileged object information. A simulator field stored beside an image is not automatically a deployable sensor observation. Future target states and label-validity masks supervise their respective branches and are excluded from the action condition. A current-object-pose oracle and a future oracle are distinct information conditions.

Physical high-level steps are separated by 50 ms. Forecast labels cover 50–400 ms relative to the source observation. Synchronous evaluation pauses the environment for policy computation, so it cannot measure the consequence of wall-clock observation age. Later fixed-delay and real-delay evaluations must retain their own time traces and report whether physics continues while the policy computes. Contact-dependent future labels require an audit of the generating action and valid prediction support; completion of that audit is **[pending for the final online-generation protocol]**.

B. Reference Implementation and Budgets

B.1. Actual Representation-Screen Configuration

The reference D model uses `se3wam_v2/split_ffn`, four transformer layers, width 768, eight attention heads, and factorized attention. DINOv3 ViT-L/16 is frozen [7]. A 224×224 agent-view image produces 196 current visual tokens. H0V has zero visual-history query tokens. The information track also supplies exact depth and camera geometry. Robot input retains eight EEF7 states at 20 Hz. Eight object tokens encode current GT pose and seven motion intervals using 25 numeric statistics and pose/interval/acceleration validity flags.

The object statistics include position, rotation vector, interval translation and relative rotation, elapsed time, linear/angular velocity, and linear/angular acceleration. Their source is simulator truth in this track. The current-pose history track removes historical object poses and all derived interval kinematics while preserving the current pose and robot history. This is a different information condition, not a different name for the same input.

D shares attention QKV/output projections and splits feed-forward networks by world/meta, flow, and action stream. Scene tokens access scene tokens; meta tokens access scene and meta; flow accesses scene and flow; action accesses meta and action. Consequently, flow supervision can change shared weights without an online flow-to-action path. The deterministic pose head pools sixteen final meta tokens. Calling the model world–action does not change these actual access relations.

B.2. Coordinates, Losses, and Generation

Target future translation is cumulative displacement in the fixed metric frame. Rotation is the left-relative vector $\phi_h = \mathrm{Log}(R_{t+\delta_h} R_t^\top)^\vee$. For valid targets $\mathcal{V}$, the pose loss is

$$\mathcal{L}_{\mathrm{pose}} = \frac{1}{|\mathcal{V}|} \sum_{h \in \mathcal{V}} \big[\|\hat{d}_h - d_h\|_2^2 + 0.1\, d_{\mathrm{SO}(3)}(\hat{R}_{t+\delta_h}, R_{t+\delta_h})^2 \big]. \tag{S2}$$

The external weight multiplying this loss is separate from the internal rotation coefficient. A full twist exponential would couple translation to rotation through its Jacobian and is not the implemented parameterization. The reported configuration does not include the former draft's heteroscedastic pose likelihood, uncertainty bridge, symmetry-minimized loss, or pose–flow consistency objective.

The action has eight E7 steps: base-frame translation increment, left-relative rotation vector, and gripper command. Translation and rotation training bounds are 0.05 m and 0.5 rad per axis. The inherited evaluation translation limit is 0.10 m per axis. Action flow matching uses a unit Gaussian in training-normalized action coordinates. Scene flow uses a metric Gaussian source with standard deviation 0.02 m. Deterministic auxiliary readouts, including pose, do not consume flow-matching source noise and time.

Only the action generator runs during Q2 deployment, using five Euler steps. Conditioning caching is on; temporal/velocity caching is off. Each replan executes one command. Trainable initialization is fresh for seed 0 rather

than inherited from a trained common-reference checkpoint. Global batch is 16. Four epochs produce 40,104 updates and 641,664 chunks. P00, P02, and P05 finish with an eight-device topology; other arms finish with sixteen. Some retained runs have recorded topology transitions. These histories preserve the declared exposure and endpoint but do not establish bitwise equivalence across topology.

B.3. Alternative Target Definitions

P04 uses a frozen Wan video VAE target. Nine RGB input frames produce three temporal latents in the target construction; this is not eight independent future latent frames. P05 uses dense cumulative 2-D displacement. P06 uses real codec block motion vectors with reference, direction, and validity. P07 uses sparse scene tracks and visibility, P08 pooled object DINO features, and P09 target tracks and visibility. P08/P09 adapt mechanisms rather than reproduce an entire published model.

Aalpha adds motion-magnitude MSE with weight 0.5 while preserving P03's Alpha-head initialization. P12, P13, and P14 add future depth, spatial DINO, and dynamic-region RGB supervision. P15 combines these three at their individual weights. A non-existent head has an inapplicable readout, not a zero error. The combination cannot identify every pairwise interaction. Raw total losses across these objectives are not commensurate.

B.4. System-Reference Training

ACT and Diffusion Policy train from scratch for 50,000 updates at global batch 32. The $\pi_{0.5}$ adaptation starts from `pi05_base` and trains 30,000 updates at global batch 256 [6]. DynamicVLA uses 50,000 updates at global batch 44. OpenWAM uses a fresh fine-tuning run from the official Alpha foundation, for five epochs, 34,490 updates at global batch 192, with its declared foundation components. DynamicWAM uses its official full initialization and 40,000 updates at global batch 128 [5, 8]. DreamVLA's 20-epoch and 40-epoch schedules use independent training runs, 48,220 and 96,440 updates at global batch 512; the shorter endpoint is not the midpoint checkpoint of the longer schedule [9].

Executors also differ: ACT uses temporal aggregation, Diffusion Policy executes eight actions, $\pi_{0.5}$ ten, and DynamicWAM sixteen before replanning. The internal representation screen executes one. System-level interpretation retains these differences. Matched-budget comparisons must also report each system's inputs, initialization, and execution configuration.

C. History Encoding and Conditional Information

C.1. Completed and Outstanding Conditions

The current-pose study compares eight history encodings plus a fresh GT-history reference. Every visual-history condition preserves current GT target pose and eight robot-state samples, while historical object pose and derived kinematics are masked. Its reference future supervision is held fixed. Completed conditions are current image, Window, Codec-MV, Flow, and PACT-Age. Flow12D, Wan, PACT-Uniform, and the fresh GT-history reference have not yet produced a complete accepted comparison table. PACT-Age distributes fifty-six history queries from older to newer frames as $[2, 4, 6, 8, 10, 12, 14]$; PACT-Uniform uses seven groups of eight. These adapt PACT-WAM's temporal allocation principle [4] through per-frame query compression, rather than reproducing its complete model.

Each of the five completed conditions has probe extraction over 350 episodes and 1,050 windows with 306 valid readouts. This establishes availability of the diagnostic arrays. The manuscript has not turned those arrays into a reported probe-effect estimate. Their final matched metrics and complete nine-condition input/noise/label audit are **[pending]**. The observed point estimates support continuing the comparison without presuming a visual-history gain.

C.2. Recipe Differences and Required Contrasts

This study uses frozen DINO-L, retains the original Alpha objective, and has a cosine schedule over the first 10,026 updates followed by three epochs at 2×10^{-5}. The combination study uses DINO-S or CNN, disables Alpha, and applies cosine scheduling across all 40,104 updates. Its stronger score therefore cannot be subtracted from the history-study score to estimate a visual-history effect.

The planned core comparison crosses current-pose versus GT-motion input with current-image versus visual-history encoding under a common recipe. Multiple training seeds and paired scenes are needed to estimate the conditional history benefit. A motion probe and an action intervention answer different questions: recoverability of velocity does not demonstrate that the policy used velocity to improve interaction.

C.3. Matched-Current and Occlusion Diagnostics

History substitutions require matching the current target pose, robot state, view, and task condition while changing valid preceding motion. The matching tolerance, group counts, and history-necessary subset acceptance remain **[pending]**. A substitute that changes the current state tests an additional intervention. Occlusion tests must report visibility and duration in physical time, while sampling-gap tests preserve the meaning of displacement by retaining elapsed

Table S2. Training exposure and development counts for all system-reference endpoints. Batch is global; all endpoints use training seed 0. Initialization and execution are part of each adapted system, not matched factors. An unspecified field requires a configuration audit before a matched-budget comparison. The two DreamVLA endpoints are independent training schedules. The 1,260-scene counts appear in the main system table.

System	Updates	Batch	Success /140	Execution / initialization note
ACT	50,000	32	50	Scratch; temporal aggregation
Diffusion Policy	50,000	32	54	Scratch; execute 8 actions
$\pi_{0.5}$	30,000	256	89	`pi05_base`; execute 10
DynamicVLA	50,000	44	24	Native continuous pipeline; initialization audit pending
OpenWAM-Alpha	34,490	192	76	Official Alpha foundation; five-epoch fine-tuning
DynamicWAM	40,000	128	72	Official full initialization; execute 16
DreamVLA: 20 epochs	48,220	512	112	Independent shorter schedule
40 epochs	96,440	512	114	Independent longer schedule
Study P00	40,104	16	88	Fresh task network; frozen DINO-L; execute 1
Study P03	40,104	16	93	Same reference recipe as P00
Study I	40,104	16	96	Fresh CNN condition; full visual history; execute 1

Table S3. Available history results. The total is a development summary, not a separate held-out test. Dashes denote unfinished comparisons.

Condition	/140	/1,260	/1,400	SR (%)
Current image	80	668	748	53.43
Window	69	593	662	47.29
Codec-MV	77	656	733	52.36
Flow	76	644	720	51.43
PACT-Age	65	569	634	45.29
Pending visual-history conditions (same information)				
Flow12D	—	—	—	—
Wan	—	—	—	—
PACT-Uniform	—	—	—	—
Pending reference with privileged target history				
Fresh GT history	—	—	—	—

time. The planned outputs link motion decoding, action change, contact outcome, and unconditional task success. No-contact failures remain in the success denominator rather than receiving zero contact error.

D. Future Supervision and Combination Selection

D.1. All Fifteen Future Conditions

All conditions in Table S4 have action supervision. Scene-flow coverage includes target regions, so P01 is not a pure context-only condition. P03 versus P01/P02 compares target-pose and scene-flow objectives under the common action path; it does not isolate an exclusive target/context spatial partition. Main-paper Fig. 3 visualizes all fifteen point estimates without adding confidence intervals absent from the source analysis.

P00/P01/P02/P03 test no future objective, scene flow, pose, and their combination. P04–P07 replace the scene target while retaining pose. P08/P09 replace the object target while retaining scene flow. Aalpha and P12–P15 add supervision to P03. Full target definitions and source units are given in Appendix B. None of these conditions changes deployment into online forecast consumption.

D.2. Paired Outcomes and Task Heterogeneity

Table S5 preserves gains, losses, and raw exact McNemar probabilities against both reference conditions. P05's adjusted probabilities are 0.00844 versus P00 and 0.01877 versus P03. The remaining positive overall differences are not established advantages. The absence of a detected difference is not an equivalence test.

Table S6 gives every task count with equal denominators. The strongest task-specific changes are exploratory. For P05 versus P03, rolling-sphere success falls from 50 to 23 while `tray_contact` failures rise from 40 to 67. Conversely, post-impact replanning rises from 54 to 60. P09's gains in translational placement and replanning coexist with fewer orientation-sensitive grasps. These observations narrow the conditions to investigate, without treating a native failure code as a learned-mechanism explanation.

The lowest count of native safety failures is P06's 250, but its 96 timeouts are the highest. P15 has only 37 timeouts and 343 safety failures. A favorable change in one category must be interpreted alongside all outcomes. Every policy failure remains in the 1,260-scene denominator.

Table S4. All fifteen future-supervision conditions, grouped by the objective changed. Each uses one training seed, four epochs, common privileged object-motion input, and action-only deployment. The 140 development scenes and 1,260-scene complement remain separate. Scene flow includes target regions.

		Development	Disjoint complement	
ID	Future supervision	Success / 140	Success / 1,260	SR (%)
Reference and component objectives				
P00	No future supervision	88	791	62.78
P01	3-D scene flow	95	782	62.06
P02	Target pose	92	789	62.62
P03	Scene flow + target pose	93	786	62.38
Scene replacements: target-pose objective retained				
P04	Wan latent + pose	92	783	62.14
P05	Dense 2-D displacement + pose	87	746	59.21
P06	Codec motion + pose	86	795	63.10
P07	Sparse scene tracks + pose	90	807	64.05
Target replacements: scene-flow objective retained				
P08	Scene flow + object DINO	89	794	63.02
P09	Scene flow + target tracks	88	808	64.13
Additional objectives: scene flow + target pose retained				
A_α	P03 + motion magnitude	89	778	61.75
P12	P03 + depth	90	791	62.78
P13	P03 + DINO	92	788	62.54
P14	P03 + dynamic RGB	89	790	62.70
P15	P03 + depth / DINO / dynamic RGB	94	802	63.65

D.3. Combination Selection

The next screen crosses three future choices, two visual encoders, and current-target RGB reconstruction on/off. It preserves a full visual window and privileged target-motion input, disables Alpha and late injection, and trains each condition fresh for four epochs with the full-run cosine schedule. Its twelve conditions are complete; matched P03/P00 controls of the selected configuration remain unfinished.

Candidate I obtains 935/1,400 by the predefined selection score. Against J, the same P07/CNN condition without reconstruction, the paired counts are 817 joint successes, 118 I-only successes, 79 J-only successes, and 386 joint failures. Its net 39-scene gain is distributed unevenly over tasks, as Table S8 shows. The selected candidate is carried forward for testing; the selection does not establish stable performance across training seeds.

D.4. Granularity and Prediction-to-Action Evidence

A future comparison must separate role coverage, variable choice, spatial allocation, and compute. The fifteen-target screen changes target packages, not only allocation at equal budget. A direct fine-target/coarse-context test still requires a compatible uniform reference with the same identity information, horizon support, and declared token/compute budget. Its result is **[pending controlled granularity comparison]**.

Physical readouts should compare target translation and rotation at each horizon; flow and tracks require valid correspondences and visibility. Their association with contact error is a separate analysis. The accepted source contains readout arrays, but this manuscript does not substitute flow-matching validation loss for freely generated physical error. The aligned prediction/contact figure requires **[generated predictions, actual states, and contact events]**. If a future oracle is introduced, it must use the same action interface and target times. Any oracle-gap ratio is reported only with a supported, non-negligible denominator.

E. Backbone, Information Flow, and Cross-Checks

E.1. D/E Backbone Comparison

BACKBONE0 is prepared but has no completed GPU training, policy evaluation, or measured-cost result. Its input is the confirmed complete history/future configuration, retaining the reference G_0 program where action does not read predicted future. D uses split feed-forward networks and shared attention projections. The adapted E organizes world and action parameters separately within the current model interface. The old native E and the object-input condition E

Table S5. Paired outcomes over the same 1,260 scenes. Gain means candidate success/reference failure; loss means candidate failure/reference success. Probabilities are exact two-sided McNemar tests. Adjustment uses the full family of fourteen P00 and thirteen P03 contrasts: $p_{\text{adj}} = \min(1, 27p)$. P03 against itself is inapplicable.

	P00: no future supervision				P03: scene flow + target pose			
ID	Gain	Loss	Raw p	Adjusted p	Gain	Loss	Raw p	Adjusted p
Reference and component objectives								
P01	80	89	0.538427	1.000000	72	76	0.805321	1.000000
P02	61	63	0.928492	1.000000	61	58	0.854630	1.000000
P03	64	69	0.728854	1.000000	—	—	—	—
Scene replacements: target-pose objective retained								
P04	74	82	0.575315	1.000000	71	74	0.868159	1.000000
P05	53	98	0.000312	0.008435	47	87	0.000695	0.018771
P06	81	77	0.811457	1.000000	68	59	0.477922	1.000000
P07	85	69	0.226651	1.000000	81	60	0.091769	1.000000
Target replacements: scene-flow objective retained								
P08	75	72	0.869050	1.000000	59	51	0.504685	1.000000
P09	91	74	0.212786	1.000000	79	57	0.071353	1.000000
Additional objectives: scene flow + target pose retained								
A_α	60	73	0.298071	1.000000	53	61	0.512255	1.000000
P12	73	73	1.000000	1.000000	63	58	0.716294	1.000000
P13	70	73	0.867250	1.000000	60	58	0.926704	1.000000
P14	71	72	1.000000	1.000000	62	58	0.784328	1.000000
P15	74	63	0.392988	1.000000	63	47	0.152338	1.000000

retain separate identities.

The planned two-arm comparison trains fresh seeds 0, 1, and 2, with the same initialization sources, data exposure, optimizer schedule, and endpoint. If the current four-epoch recipe is retained, there are 240,624 updates across six runs and 8,400 evaluation episodes at 1,400 per endpoint. These are planned counts. The active-backbone parameter difference is targeted within 5%; frozen frontend parameters and their runtime costs are recorded separately. Matching exposure or parameter count does not match training FLOPs.

The predeclared practical SR and non-inferiority margins are two percentage points, with a 10% p95 latency benefit considered for cost selection. Intervals must support the corresponding superiority, equivalence, or non-inferiority criterion before that label is used. If three seeds are inconclusive, both arms may be expanded to five under the existing plan. Same-device public-policy timing includes internal perception/action computation and excludes external rendering and oracle access. Selection and uncertainty are **[pending]**.

E.2. Seven Information-Flow Conditions

R0 versus R1 tests auxiliary supervision under action-only deployment. R1's two deployment modes test the cost and effect of generating an unused future from the same checkpoint. R1–R4 examine information access. R2 versus R5 and R3 versus R6 compare synchronous and sequential programs. Sequential second stages train on clean first-stage labels and deploy on first-stage predictions. The corresponding train/test mismatch belongs in the interpretation.

The current generative state is scene flow with shape $[B, S, 196, 8, 3]$; pose, tracks, depth, DINO, and RGB can be deterministic auxiliary readouts. A selected sparse-track or Wan representation requires its own generated-state, normalization, noise, and loss definition before this experiment can run. Replacing it with scene flow would change the question. This interface is **[pending final representation binding]**.

The planned three seeds yield twenty-one trained configurations and twenty-four deployment configurations, including R1's second mode; at 1,400 scenes each, that is 33,600 rollout episodes. Alpha is fixed to zero. Per-modality Euler5 is common, but joint inference uses five combined calls and sequential inference uses five plus five. Actual calls, latency, parameters, training exposure, and valid readouts are required alongside SR. The planned counts do not imply completed runs.

E.3. Mechanisms and Final Design

An independently replaceable future interface permits oracle, valid matched-current swap, time-shift, or calibrated-noise diagnostics. Each intervention holds perception, robot state, and action-source noise fixed. Changes in action alone show

Table S6. Future-study successes: ninety scenes per task and 1,260 per condition. Columns follow the task families in Table S1; all tasks are retained, including zero-success tasks. Occl. denotes physical occlusion; Trans. denotes translating-receptacle placement; Tabletop denotes tabletop sphere capture.

	Reference	T1: target motion				T2: placement		T3: passage		T4: capture				T5	
ID	Grasp	Belt	Occl.	SO(3)	XYZ	SE(3)	Trans.	Full	Phase	Can	Slider	Sphere	Tabletop	Replan	Total
Reference and component objectives															
P00	85	90	90	75	76	0	8	3	15	90	69	45	86	59	791
P01	84	90	90	64	69	0	18	0	11	90	74	46	90	56	782
P02	87	90	90	78	73	0	7	0	0	90	71	59	89	55	789
P03	86	90	88	75	77	0	7	0	6	86	80	50	87	54	786
Scene replacements: target-pose objective retained															
P04	83	85	84	73	76	0	20	0	0	90	72	54	89	57	783
P05	84	89	90	66	75	0	10	0	0	90	71	23	88	60	746
P06	86	88	89	79	70	0	6	4	2	89	72	60	85	65	795
P07	82	90	90	78	71	0	17	0	0	89	79	55	90	66	807
Target replacements: scene-flow objective retained															
P08	84	90	90	71	81	0	13	0	0	90	77	60	90	48	794
P09	84	90	89	65	78	0	21	0	0	90	77	59	90	65	808
Additional objectives: scene flow + target pose retained															
A_α	86	90	90	72	75	0	3	0	0	85	73	55	90	59	778
P12	83	90	90	76	76	0	5	0	0	89	77	57	90	58	791
P13	86	90	90	80	73	0	6	0	0	88	78	54	90	53	788
P14	85	89	90	72	79	0	5	0	0	90	80	53	88	59	790
P15	87	90	90	79	81	0	11	0	0	90	75	54	90	55	802

Table S7. Native evaluator failure counts out of 1,260 scenes per condition, in the same design groups as the success tables. Safety is the evaluator category, not a certified safety measure. Every failure remains in the success-rate denominator.

ID	Future supervision	Boundary	Safety	Task viol.	Timeout	Total
Reference and component objectives						
P00	No future supervision	84	297	17	71	469
P01	3-D scene flow	62	324	23	69	478
P02	Target pose	77	331	17	46	471
P03	Scene flow + target pose	76	295	51	52	474
Scene replacements: target-pose objective retained						
P04	Wan latent + pose	79	310	41	47	477
P05	Dense 2-D displacement + pose	87	356	14	57	514
P06	Codec motion + pose	96	250	23	96	465
P07	Sparse scene tracks + pose	47	329	20	57	453
Target replacements: scene-flow objective retained						
P08	Scene flow + object DINO	61	322	42	41	466
P09	Scene flow + target tracks	67	299	35	51	452
Additional objectives: scene flow + target pose retained						
A_α	P03 + motion magnitude	57	315	65	45	482
P12	P03 + depth	62	327	40	40	469
P13	P03 + DINO	71	326	27	48	472
P14	P03 + dynamic RGB	70	322	18	60	470
P15	P03 + depth / DINO / dynamic RGB	59	343	19	37	458

Table S8. Combination-study successes, grouped by future target and visual encoder. DINO is frozen DINO-S; CNN is trained from scratch. Minus/plus denotes current-target RGB reconstruction off/on; parentheses retain source condition IDs. Each task has one hundred scenes, including development scenes, and each condition has one training seed. Totals use the predefined 1,400-scene selection pool.

Task	P15: additional targets				P07: scene tracks				P09: target tracks			
	DINO− (B)	DINO+ (A)	CNN− (D)	CNN+ (C)	DINO− (H)	DINO+ (G)	CNN− (J)	CNN+ (I)	DINO− (L)	DINO+ (K)	CNN− (N)	CNN+ (M)
Reference grasp												
Grasp	90	89	91	92	90	92	87	90	79	85	77	87
T1: target motion												
Belt	100	100	99	100	100	100	100	97	99	100	90	99
Occlusion	100	100	99	99	99	100	100	97	96	99	82	98
SO(3)	86	71	80	88	75	87	69	86	69	62	64	78
XYZ	58	58	82	75	73	75	89	82	44	56	82	84
T2: receptacle motion												
SE(3)	0	0	0	0	0	0	0	0	0	0	0	0
Translation	8	4	6	5	9	1	14	4	4	5	9	14
T3: constrained passage												
Full	0	0	0	0	0	0	0	0	0	0	0	0
Phase	0	0	0	0	0	0	28	72	0	0	0	2
T4: object capture												
Can	99	99	99	98	100	99	99	100	97	99	96	99
Slider	88	85	88	91	74	83	86	81	84	84	88	88
Sphere	71	67	67	63	47	40	53	56	68	68	62	64
Tabletop sphere	100	100	100	99	99	100	95	98	100	100	100	100
T5: post-impact replanning												
Replan	73	63	64	57	74	72	76	72	80	72	79	57
Total / 1,400	873	836	875	867	840	849	896	935	820	830	829	870

Table S9. Planned Q3 conditions; no measured result is assigned to these rows. F denotes a generated future state and A an action state. Arrows specify information access, independently of whether the future loss is enabled. R1 has two deployments of the same trained checkpoint. Sequential second stages use clean ground truth in training and generated first-stage values at test time.

ID	Future loss	$F \to A$	$A \to F$	Cross-stream training input	Inference order
R0	No	—	—	None	A only
R1	Yes	—	—	None	A only
				Same checkpoint	Joint F and A
R2	Yes	Yes	—	Synchronous noisy states	Joint F and A
R3	Yes	—	Yes	Synchronous noisy states	Joint F and A
R4	Yes	Yes	Yes	Synchronous noisy states	Joint F and A
R5	Yes	Yes	—	Clean F for second-stage A	F, then A
R6	Yes	—	Yes	Clean A for second-stage F	A, then F

sensitivity; evidence for useful forecast use also requires task and safety outcomes. Removing a branch at inference and retraining without its loss answer different questions.

After selecting information flow, the study revisits retained future representations and applies the selected flow to the unselected backbone. History choices are revisited when the future target changes. Only the necessary comparisons are repeated; previously valid matched endpoints can be reused. Ranking reversals change the conditional design choice. Multi-seed and scenes not used for selection are required before passing a complete (H, Z, B, G) design to pretraining. The selected configuration, cross-check estimates, and mechanism readouts remain **[pending]**.

F. Pretraining, Cost, and External Evaluation

F.1. Measured Policy Cost and Physical Delay

The future-screen policy call means span 177.071–184.212 ms; p95 spans 178.726–185.133 ms; peak allocated memory spans 1.571–1.617 GiB. These are same-device, fixed-order measurements including internal perception and action computation. Loading, external oracle reads, I/O, rendering, and priming are excluded. One measurement round supports this range, not a robust ranking of differences of a few milliseconds.

For deployed commands, action age is the difference between sensor capture and command application. The planned deployment profile separates capture, availability, preprocessing, inference, queuing, issue, and application. It records hardware, precision, batch size, warm-up, synchronization, and timed sample counts. Fixed-delay simulation tests a controlled lag; wall-clock execution tests the actual combined system. The final same-device profiles, deadline misses, and action-age distributions are **[pending]**. A policy's public function latency must not be labeled full environment end-to-end latency when parts of that path are excluded.

F.2. Recipe and Robot-Data Base

After the design cross-checks, the study fixes input modalities, history support, future targets/horizons, information flow, and action semantics. The next stage optimizes the existing loss weights and training recipe, and changes depth/width within the chosen architecture for scale comparison. Changing the representation or adding an information path belongs to a new design comparison rather than an optimizer ablation.

DROID supplies real robot demonstrations [2]. Preparation checks sensor timing, calibration, depth, command versus measured motion, gripper semantics, and training-only normalization. It does not provide object pose GT; the target-pose pipeline must be separately validated for any selected pose-based objective. Only targets used by the selected design and necessary ablations need to be produced. Valid label masks and rejected-action rates are part of the data report, not silently filled values. This stage is **[planned]**.

The formal base is trained on real robot data. Two independent post-training branches start from the same base: one for simulation and one for real-robot tasks. Simulation-from-scratch experiments supply recipe experience and a strong reference, while their weights do not initialize the formal robot-data base or the real-robot branch. Comparisons with and without pretraining share downstream data and exposure, and additionally allow equally budgeted recipe searches. They report the pretraining cost explicitly. Data, model, and compute scale have separate axes; unrun settings are not points on a scaling curve.

Table S10. Evidence slots for the final model. All are planned and currently unmeasured in this manuscript.

Track	Evidence question and required output
Unused simulation scenes	Does selection generalize? Multi-seed SR and uncertainty on unused scenes.
Robot-data base	Does initialization help? Paired adaptation, data/compute scale, and cost.
DOMINO single arm	Does the design transfer to other dynamic tasks? Native score and adaptation details.
Static compatibility	Are static skills retained? Native score with common tuning.
Real conveyor	Does the integrated system work physically? Trial counts, success, and interventions.
Measured delay	Does performance survive actual action age? Timing, safety, and success.

F.3. Planned External and Real-Robot Evidence

Single-arm DOMINO [1] tests dynamics under a different task/data protocol. Static compatibility uses an explicitly selected native suite such as LIBERO [3]. Real conveyor manipulation evaluates perception, prediction, action, and timing together. The final phase reuses completed pretraining, scaling, and independent adaptation results, then fills uncovered object, scene, motion, and delay conditions. It does not repeat an already completed scaling matrix.

The real-robot protocol must specify objects, speed/delay strata, trial allocation, ordering, calibration, safety termination, and human intervention. The old budget of 180 trials is not a completed count or an adopted current sample size. External native scores are not pooled into the fourteen-task simulation SR. Final training scale, robot trials, and real-world performance are **[pending]**.

G. Failure Evidence and Qualitative Experiments

G.1. Supervision at the Executed Action Position

The hard-task investigation finds that all 800 historical full-passage demonstrations release only at the final action. Release is represented at chunk position h_7 but has no h_0 supervision in that set, whereas the deployment program executes h_0. Thus the problem is action-position coverage. It cannot be summarized as completely missing release labels.

Short tail-window training produces 2/16 successes in one diagnostic validation and 1/6 in a follow-up selected-scene check; this does not establish reliable full-task learning. For phase passage, three seeds after 256 updates give 0/96, 12/96, and 26/96; at 1,024 updates all three give 0/96. Six first-action-weighting endpoints total 576 failures in 576 trials, and six GT-role diagnostic endpoints also total zero successes in 576. These are local diagnostic budgets, sepa-

rate from the fourteen-task study.

A zero action-source deployment intervention changes the three 256-update scores from 0/12/26 to 0/1/46 successes out of 96 each. It was selected after observing existing outcomes and is exploratory. A reproducibility investigation attributes some same-seed trajectory branching to RGB channel changes of one gray level; fixed full-input/RNG inference and fixed-action physical replay remain consistent. This finding does not explain every historical trajectory difference.

G.2. Gripper Execution Diagnostics

The gripper investigation compares sign decoding, a fixed 0.1 dead zone, and nearest-label ternary projection and history-hold mappings. The ternary projection maps to $\{-1, 0, +1\}$ with thresholds at ± 0.5. For the reported fifth-epoch endpoint, the 1,260-scene counts are 698 for sign, 742 for the fixed dead zone, 749 for nearest-label ternary projection, and 741 for history hold. The ternary projection does not show a clear overall improvement over the fixed dead zone; history hold adds failures outside the capture subset. Neither is adopted as the production default in the evidence cited here.

The broader fourth-to-eighth-epoch dead-zone report contains descriptive counts but retains an unsuccessful strict historical trajectory-consistency audit. It is not used as an unrestricted main-table performance claim. These diagnostics show why a decoder change must be distinguished from a representation change and why both recovered and newly failed episodes need to be reported.

G.3. Figures and Aligned Rollouts

Main-paper Fig. 1 illustrates interaction geometry conceptually. Fig. 2 depicts the implemented D reference with P03's auxiliary targets. Fig. 3 plots measured success-rate differences and paired scene outcomes. The actual-scene task montage from the earlier benchmark assets can be used to illustrate the suite after its release identity is checked.

The remaining qualitative comparison aligns source observation, predicted future, actual future, and the contact event at the same physical times. It should include P05's adverse sphere condition, a retained sparse-trajectory condition, and a matched future-to-action intervention after that interface exists. Each case records method, scene, camera, frame, units, visibility, action application, and outcome. These assets are **[pending synchronized prediction/rollout exports]**.

Case selection must include the failures that alter the main conclusion. A successful clip illustrates an accepted outcome; it does not replace the full success denominator. Anonymous supplementary media should report filename, episode, duration, playback rate, cuts, resets, and human interventions. No real-robot video or forecast intervention video is claimed as delivered by the present manuscript.

References

[1] Heng Fang, Shangru Li, Shuhan Wang, Xuanyang Xi, Dingkang Liang, and Xiang Bai. Towards generalizable robotic manipulation in dynamic environments. In *European Conference on Computer Vision*, 2026. 9

[2] Alexander Khazatsky, Karl Pertsch, Suraj Nair, et al. DROID: A large-scale in-the-wild robot manipulation dataset. *arXiv preprint arXiv:2403.12945*, 2024. 9

[3] Bo Liu, Yifeng Zhu, Chongkai Gao, Yihao Feng, Qiang Liu, Yuke Zhu, and Peter Stone. LIBERO: Benchmarking knowledge transfer for lifelong robot learning. *arXiv preprint arXiv:2306.03310*, 2023. 9

[4] Yushan Liu, Jingjing Fan, Shoujie Li, Yifan Xie, Xiao-Ping Zhang, and Wenbo Ding. PACT-WAM: Predicting Actions and Visual Foresight with Compact Temporal Encoding for Robot Manipulation. *arXiv preprint arXiv:2602.15882*, 2026. 3

[5] Yunfan Lou, Hewen Gao, Xiyu Zhu, Zhuoran Qiao, Xuan Han, Yifan Yang, Yifan Ye, Boxian Yao, and Zhibo Pang. DynamicWAM: Dual-path motion conditioning for world-action models in dynamic manipulation. *arXiv preprint arXiv:2608.00793*, 2026. 3

[6] Physical Intelligence, Kevin Black, Noah Brown, James Darpinian, et al. $\pi_{0.5}$: a vision-language-action model with open-world generalization. *arXiv preprint arXiv:2504.16054*, 2025. 3

[7] Oriane Siméoni, Huy V. Vo, Maximilian Seitzer, Federico Baldassarre, Maxime Oquab, Cijo Jose, Vasil Khalidov, Marc Szafraniec, Seungeun Yi, Michaël Ramamonjisoa, et al. DINOv3. *arXiv preprint arXiv:2508.10104*, 2025. 2

[8] Yuran Wang, Siqiao Huang, Mingleyang Li, Chenhao Zhang, Jiaqi Liang, Weiyang Jin, Yue Chen, Xuemin Chi, Donghao Zhou, Qize Yu, Yu-Kai Wang, Yuhan Rui, Shenzhe Yao, Zhen Yuan, Zhenhao Shen, Kefei Zhu, Zijie Zhu, Ning Gao, Xiaowei Chi, Guanqi He, Shanghang Zhang, Hao Dong, Lin Shao, and Hang Zhao. OpenWAM: An Open, Modular Exploration Towards Systematic World-Action Model Pretraining. *arXiv preprint arXiv:2609.07398*, 2026. 3

[9] Wenyao Zhang, Hongsi Liu, Zekun Qi, Yunnan Wang, Xin-Qiang Yu, Jiazhao Zhang, Runpei Dong, Jiawei He, He Wang, Zhizheng Zhang, Li Yi, Wenjun Zeng, and Xin Jin. DreamVLA: A vision-language-action model dreamed with comprehensive world knowledge. In *Advances in Neural Information Processing Systems*, pages 24195–24228, 2025. 3